



The Capacity That Never Drops

Auditing the bike-share helmet bin's RFID return log against a term of manually counted helmets

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Background

Riders report empty helmet bins by evening even as the bike-share scheme's RFID return log holds near full capacity. The sensor confirms a tap at the slot; it has never confirmed a helmet landed inside it.

Aims

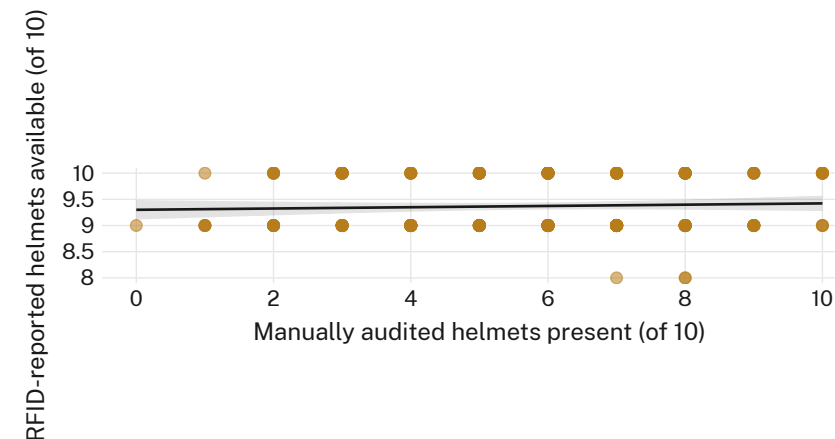
- Compare weekly audited helmet counts against the RFID system's own reported availability
- Characterise how riders satisfy the return sensor without depositing a helmet
- Test whether the audited-reported gap widens as tap volume grows

Methods

- 18 docks · weekly hand audit · 12-week term (n = 216 dock-weeks, Apr-Jul 2026)
- RFID return-tap log cross-referenced to audit timestamps
- Sensor's 2 m tap-detection radius held constant throughout

Results

Audited counts fell from a dock mean of 7.6 helmets in week 1 to 3.4 in week 12 (of a 10-helmet capacity); the RFID-reported count held between 8 and 10 throughout, tracking the audited count only weakly ($r = 0.05$).



Across 216 dock-weeks between April and July 2026, the RFID system's reported helmet count held between 8 and 10 of a 10-helmet capacity regardless of what a same-week hand audit found inside the bin ($r = 0.05$).

Discussion & conclusions

The sensor logs a tap, not a deposit — riders learned the difference within the term's first fortnight. Facilities' capacity-planning dashboard still schedules restocking from the RFID figure. Next: pair the tap sensor with the load cell already piloted on two docks.

References

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- Helmet identifiers logged by dock only; no rider identity retained. Ethics exemption 2026/061.

